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The role of carrier spectral composition in the perception of musical pitch

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Abstract

Temporal envelope fluctuations of natural sounds convey critical information to speech and music processing. In particular, musical pitch perception is assumed to be primarily underlined by temporal envelope encoding. While increasing evidence demonstrates the importance of carrier fine structure to complex pitch perception, how carrier spectral information affects musical pitch perception is less clear. Here, transposed tones designed to convey identical envelope information across different carriers were used to assess the effects of carrier spectral composition to pitch discrimination and musical-interval and melody identifications. Results showed that pitch discrimination thresholds became lower (better) with increasing carrier frequencies from 1k to 10k Hz, with performance comparable to that of pure sinusoids. Musical interval and melody defined by the periodicity of sine- or harmonic complex envelopes across carriers were identified with greater than 85% accuracy even on a 10k-Hz carrier. Moreover, enhanced interval and melody identification performance was observed with increasing carrier frequency up to 6k Hz. Findings suggest a perceptual enhancement of temporal envelope information with increasing carrier spectral region in musical pitch processing, at least for frequencies up to 6k Hz. For carriers in the extended high-frequency region (8–20k Hz), the use of temporal envelope information to music pitch processing may vary depending on task requirement. Collectively, these results implicate the fidelity of temporal envelope information to musical pitch perception is more pronounced than previously considered, with ecological implications.

Keywords Temporal envelope · Musical pitch perception · Carrier · Transposed tone · Pitch interval · Melody identification

Introduction

Naturally occurring sounds contain dynamically varying temporal envelope modulations. The envelope information of sounds contributes to facilitate many ecologically relevant auditory functions ranging from sound localization, music, and speech perception to conspecific communication (Moore, 2019; Oxenham, 2018). Musical pitch processing, in particular, has been thought to rely primarily on encoding the temporal periodicity of a sound waveform through phase-locked responses of auditory nerve (AN) fibers. Supporting evidence comes from studies showing that performance on musical interval discrimination, melody perception, and absolute-pitch identification degrades substantially for tones with frequencies above 4–5k Hz (Hatoh & Ohgushi, 1991; Moore, 2019; Semal & Demany, 1990), where phase-locking cues become unavailable. Similarly, imposing temporal envelope modulations on aperiodic complexes such as sinusoidally amplitude-modulated (SAM) noise (i.e., noise for which spectral cues are unavailable) can evoke a salient pitch percept allowing melodic and interval identification (Burns & Viemeister, 1976, 1981). The observation that the highest pitch produced by most musical instruments falls within 5k-Hz frequency range may also play a role in temporal based accounts for musical pitch perception (Plack & Oxenham, 2005a, b). Taken together, these findings suggest temporal envelope modulations provides robust frequency information in facilitating the perception of musical pitch.

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From the perspective of signal composition, musical sounds, like all complex tones, can be decomposed into two constituent components: a slowly varying temporal envelope and a rapidly oscillating carrier. While the importance of temporal envelope modulation has been demonstrated, relatively few studies have examined the impact of carrier fine structure on musical pitch processing. In fact, psychophysical findings have demonstrated that pitch derived from envelope periodicity is affected by the presence of the conveying carrier fine structure information of a complex tone. One line of evidence comes from the pitch shift phenomenon: harmonic complexes with a frequency shift of each component generate the perception of a pitch shift while preserving identical temporal envelopes (de Boer, 1956; Schouten et al., 1962). Prior research has found that listeners can successfully discriminate between pitches of harmonic and frequency-shifted complex tones based on carrier fine structure cues, even when bandpass-filtered to the same spectral region to limit the potential use of resolved harmonics (Hopkins & Moore, 2009; Moore et al., 2006; Moore & Sęk, 2009; Moore & Moore, 2003). The effects of carrier fine structure on pitch encoding have also been demonstrated in studies that reported pitch-matching responses for transposed single tones comprising inharmonic components to correspond better to carrier periodicity, rather than envelope periodicity (Grimault, 2016; Santurette & Dau, 2011). These findings provide evidence that carrier fine structure affects pitch extraction based on the temporal envelope of complex tones.

Despite the demonstrated importance of carrier fine structure in complex pitch perception, precisely how fine-structure information affects musical pitch perception remains less well-understood. The goal of this study was to examine the extent to which carrier spectral information impacts the perception of musical pitch conveyed by temporal envelope modulations. To address this, we used a stimulus design known as transposed tones (van de Par & Kohlrausch, 1997; Luo et al., 2014; McClaskey, 2017) such that the temporal patterns in the modulated envelopes of high-frequency carriers are comparable to the temporal response pattern of its low-frequency counterparts. This permitted superior assessment of the potential interfering effects of carrier spectral composition to musical pitch perception under conditions of identical temporal envelope information. We evaluated carrier frequencies both below and above the putative limits assumed for phase-locking in humans (4–5k Hz; Verschooten et al., 2019). For transposed tones with carrier frequencies below 4k Hz, the temporal phase-locking cues could be contributed by the modulating envelopes and/or by the concurrent carrier fine structure. As such, contributions from spectral and temporal envelope information could not be completely disentangled (Kale et al., 2014; Moore & Moore, 2003). In comparison, for transposed tones with carrier frequencies above 4k Hz, temporal phase-locking cues is assumed to be contributed primarily by the modulating envelope and could be separated from its spectral information (Oxenham et al., 2004). Based on these differences, temporal envelopes placed on lower- versus higher-frequency (>4k Hz) carriers should be differentially affected in providing information to musical pitch perception.

In three experiments, we investigate the ability of listeners to discriminate fundamental frequency (F0) differences, compare musical pitch intervals, and identify melodies conveyed in the envelope modulations of SAM or complex harmonics transposed onto sinusoidal-carriers spanning 1k, 4k, 6k, and 10k Hz, or a noise carrier, as well as presented in isolation. An envelope consisting of three-harmonics to create the missing F0 condition was included in order to examine whether pitch perception could be further strengthened by the envelope of complex-harmonics, which has been shown to generate stochastic neural firing patterns conducive to better performance than pure sinusoidal envelopes (i.e., SAM; Penninger et al., 2013; Rubinstein et al., 1999). We hypothesize that the use of temporal envelope information to derive musical pitch should be differentially influenced by the spectral region of conveying carriers. We report a surprising enhancement to the perception of musical pitch induced by carrier spectral region, with better sensitivity to temporal envelope cues with increasing carrier frequency up to 6k Hz. For carriers belonging to the extended high-frequency region (EHF: 8–20k Hz), the perceptual enhancement of temporal envelope information to musical pitch processing varies depending on task requirement. Findings suggest that the fidelity of temporal envelope information to musical pitch perception may be more pronounced than previously considered, at least for carrier frequencies up to 6k Hz.

Experiment 1: F0 difference limen (F0DL) for the pitch of transposed tones

Experiment 1 examined the effects of carrier spectral region on F0DLs in pitch discrimination of transposed tones modulated with either a SAM or a three-harmonic complex envelope of identical F0s. A two-interval F0 discrimination task with adaptive tracking was used to measure F0DLs. If carrier spectral region affects envelope periodicity detection, listeners should produce higher, or unattainable, thresholds in frequency discrimination as carrier frequency increases.

Method

Participants

A total of 16 participants (14 females) with a mean age of 23.0 years (*SD* = 2.28) years participated. All participants were native speakers of Mandarin Chinese and right-handed as identified using the Edinburgh Handedness Inventory–Short Form version (Oldfield, 1971; Veale, 2013). All participants had normal hearing to within 20 dB HL at octave frequencies 250 Hz to 8000 Hz assessed with standard audiometric procedure (Maico MA25; MAICO, Minneapolis, MN, USA). Participants reported no history of audiological or neurological disease. None of the participants had any prior experience in psychophysical experiments. All participants completed the Montreal Music History Questionnaire (MMHQ; Mandarin version, Coffey et al., 2011). All experiment protocols were approved and performed in accordance with the guidelines of the Research Ethics Committee at National Taiwan University, Taiwan. Prior to the experiment, written informed consent was obtained from each participant in accordance with the Research Ethics Committee at National Taiwan University, Taiwan.

Stimuli and apparatus

Stimuli were generated using MATLAB software (The MathWorks, Natick, MA) on an ASUS Vento PC and presented at a sampling rate of 44.1 k Hz through 16-bit digital-to-analog converters (Creative Sound Blaster X-Fi Titanium). Each tone complex was generated by multiplying either a SAM envelope or an envelope made by the addition of the 3rd to 5th harmonics of the same F0 (i.e., missing- fundamental harmonic complex) with a sinusoidal frequency carrier or a low-pass noise carrier. The complex tone with sinusoidal envelope and carrier can be represented using the following formula,

$$AM(t) = sin\left(2\pi f_c(t) + \phi_c\right)\left(1 + cos\left(2\pi f_m(t) + \phi_m\right)\right), \tag{1}$$

where t represents time, f_c and f_m were the carrier and modulation frequencies respectively, and ϕ_c and ϕ_m were their respective phases. In this experiment, the phase of the carrier and modulators were both set to 0. Two types of AM-envelope conditions were generated. In the SAM envelope condition, the sinusoidal tone was used as the AM envelope and transposed onto five different carrier types: 1k, 4k, 6k, or 10k Hz and a low-pass noise at 12k Hz. In the missing-F0 AM-envelope condition, the 3rd, 4th, and 5th harmonics of the original sinusoidal tone were added together to create an equivalent modulating envelope of the missing-F0 pitch. The F0-envelope was then transposed onto the same four types of frequency carriers: 1k, 4k, 6k, 10k Hz, and a low-pass noise (12k Hz cutoff), yielding a total of 10 stimuli conditions (2 envelope types × 5 carriers). A control condition with no carrier (i.e., envelope only) was included for comparison. Before multiplication, the envelope was half-wave rectified and low-passed with a Butterworth fourth order filter at 0.2* f_c (where f_c is the carrier frequency) to approximate auditory peripheral representations (Oxenham et al., 2004). Figure 1 illustrates the procedure for generating the SAM or missing-F0 envelope-modulated transposed tone.

All stimuli were presented through Sennheiser headphones (HD380 Pro) in a double-walled, acoustically isolated steel chamber (interior dimensions 2.5 m × 2.5 m × 2 m; Industrial Acoustics Company). All stimuli had linear rise-decay ramps each lasting 5% of the overall duration. Stimulus levels were calibrated to 70 dBA using a 6-cc coupler, 0.5-inch microphone (Bruel & Kjaer, Model 4189), and a Precision Sound Analyzer (Bruel & Kjaer, Model 2250).

Procedure

The experiment was conducted in a random block design with a total of 12 blocks (2 envelope type × 6 carriers). In a given block, one combination of envelope modulation (SAM or missing-F0 harmonic complex) and carrier type (none, 1k, 4k, 6k, 10k Hz and low-pass noise at 12k Hz) was randomly selected to produce the transposed tones. A two-alternative forced-choice design was employed to test the ability of listeners to discriminate small frequency differences of a tone pair. An adaptive two-down one-up (2D1U) staircase procedure was used to track participant's 70.7% threshold for psychometric function (Levitt, 1971). On each trial, one pair of modulated tones each at 500-ms duration were presented, in which the standard interval contained the transposed stimuli modulated with an envelope F0 at 110 Hz, and the comparison interval contained the same type of stimuli with envelope modulation at a different frequency. The starting frequency of the comparison interval was initialized at 155.56 Hz (i.e., 1/2 octave higher). Each tone was gated on and off with a 20-ms rise and decay cosine ramps. The two intervals were separated by a 200-ms silent ISI. The nominal reference frequency of 110 Hz was roved by 10% on each trial. The presentation order of the two intervals was randomized on each trial.

At each step, Δi will be increased or decreased by a ratio of 0.1 (Δi/i) up to and including the 4th reversal, and 0.01 thereafter. Each run of the adaptive track consisted of 50 trials. The participant was required to identify which of the two intervals contained the perception of a higher pitch by pressing the number "1" key if the first interval contain a higher pitch, or the number "2" key if the second interval contain a higher pitch. Visual feedback was provided after each trial for adaptive tracking. All stimuli were presented in a continuous background of band-pass filtered pink noise (i.e., noise between 30 and 1000 Hz with greater power in the lower-frequency portion of the spectrum) at 57 dB SPL. Each threshold estimate was based on the mean of the last four to six reversals of the adaptive track. Each participant completed a total of four tracks of 50 trials/track for each of the 10 blocks, giving a total of 2,400 trials (50 trials × 4 adaptive tracks × 12 blocks).

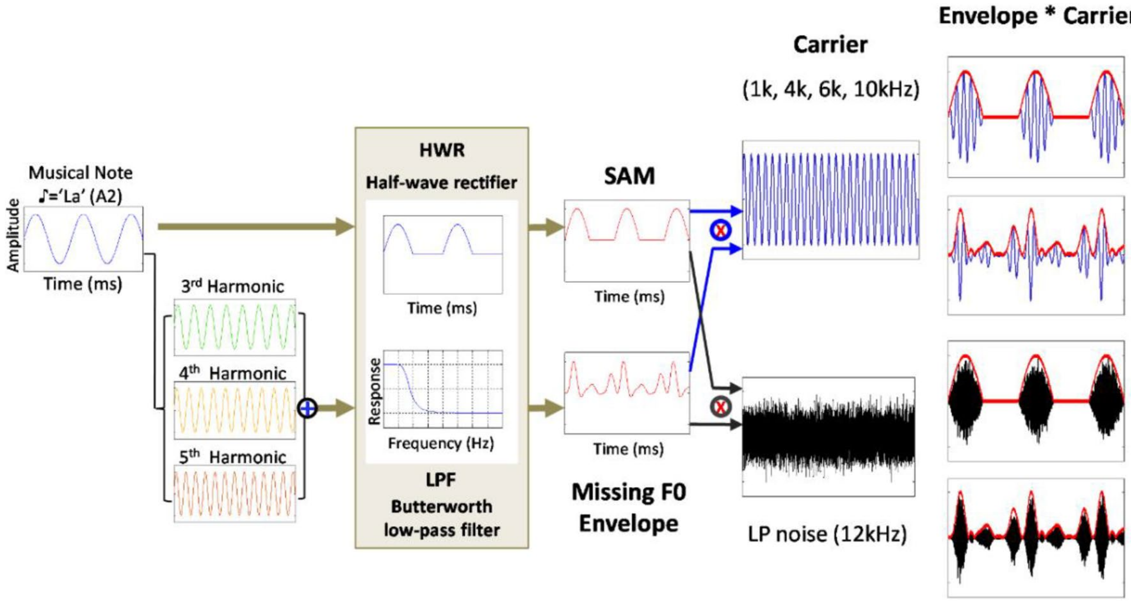

Fig. 1 Stimulus generation procedure for the transposed tone stimuli in Experiments 1 to 3. Each transposed tone comprised either a SAM envelope or an envelope made by the addition of the 3rd to 5th harmonic components with the same musical note frequency and multiplied onto sinusoidal carriers (1k, 4k, 6k, and 10k Hz) or a low-pass filtered noise with cutoff frequency at 12k Hz. The envelope is half-wave rectified and low passed with a 4th order Butterworth filter before multiplication with the carrier. (Color figure online)

Data analysis

All statistical analyses were performed using SPSS 18.0 and custom written programs in MATLAB (The MathWorks, Natick, MA). An alpha level of $p < .05$ (two-tailed) was selected a priori to determine statistical significance. Standard parametric tests were performed at the group level (repeated-measures ANOVAs, paired-samples t tests, and Pearson product-moment correlations). Multiple comparisons were corrected using Bonferroni corrections (familywise $\alpha = 0.05$). Responses were recorded and computed as thresholds or response accuracy according to task requirements using MATLAB scripts. Pitch thresholds were estimated in percent difference by averaging across the last 4–6 reversal points on the adaptive track for each participant. All analyses were then performed on the log-transformed F0DLs, defined as the geometric mean across 4 repetition of tracks per condition per subject.

Results

Carrier type modulates pitch discrimination thresholds of transposed tones

Figure 2 summarizes the results for the F0-pitch threshold tracking task. A two-way repeated-measures ANOVA on F0-pitch thresholds with the factors envelope (SAM vs. three-harmonic) and carrier type (none, noise, 1k, 4k, 6k, and 10k Hz) revealed a significant difference of pitch thresholds obtained for transposed tones modulated with SAM or three-harmonic envelopes, $F(1, 14) = 7.440$, $p = .016$ (Fig. 2a). There was a significant main effect of carrier type, $F(5, 70) = 13.147$, $p < .001$ (Fig. 2b). No significant interaction between carrier type and envelope was obtained, $F(5, 70) = 0.545$, $p = .742$. Bonferroni-corrected post hoc comparisons of pitch thresholds by carrier type indicated that the threshold was significantly lower when there was no carrier (envelope only) compared with the 1k Hz or noise carrier, $t(15) = -2.657$, $p = .018$ and $t(15) = -5.006$, $p < .001$, respectively. The only other significant pair-wise comparison was between the pitch thresholds for discriminating transposed tones on higher-frequency carriers (6k, 10k Hz) and noise carrier, with noise carrier producing higher thresholds—6k: $t(15) = -4.414$, $p = .001$; 10k: $t(15) = -3.805$, $p = .002$. All other pair-wise comparisons were not significant.

Musical experience on pitch discrimination with transposed tones

To assess whether music training experience may benefit F0 pitch discrimination for transposed tones, participants were further divided into musician and nonmusician groups according to the following criteria (data collected from MMHQ). For the musician group, qualifying participants were required to have commenced formal music training before the age of 6 (mean age = 5.61 years, SD = 1.85)

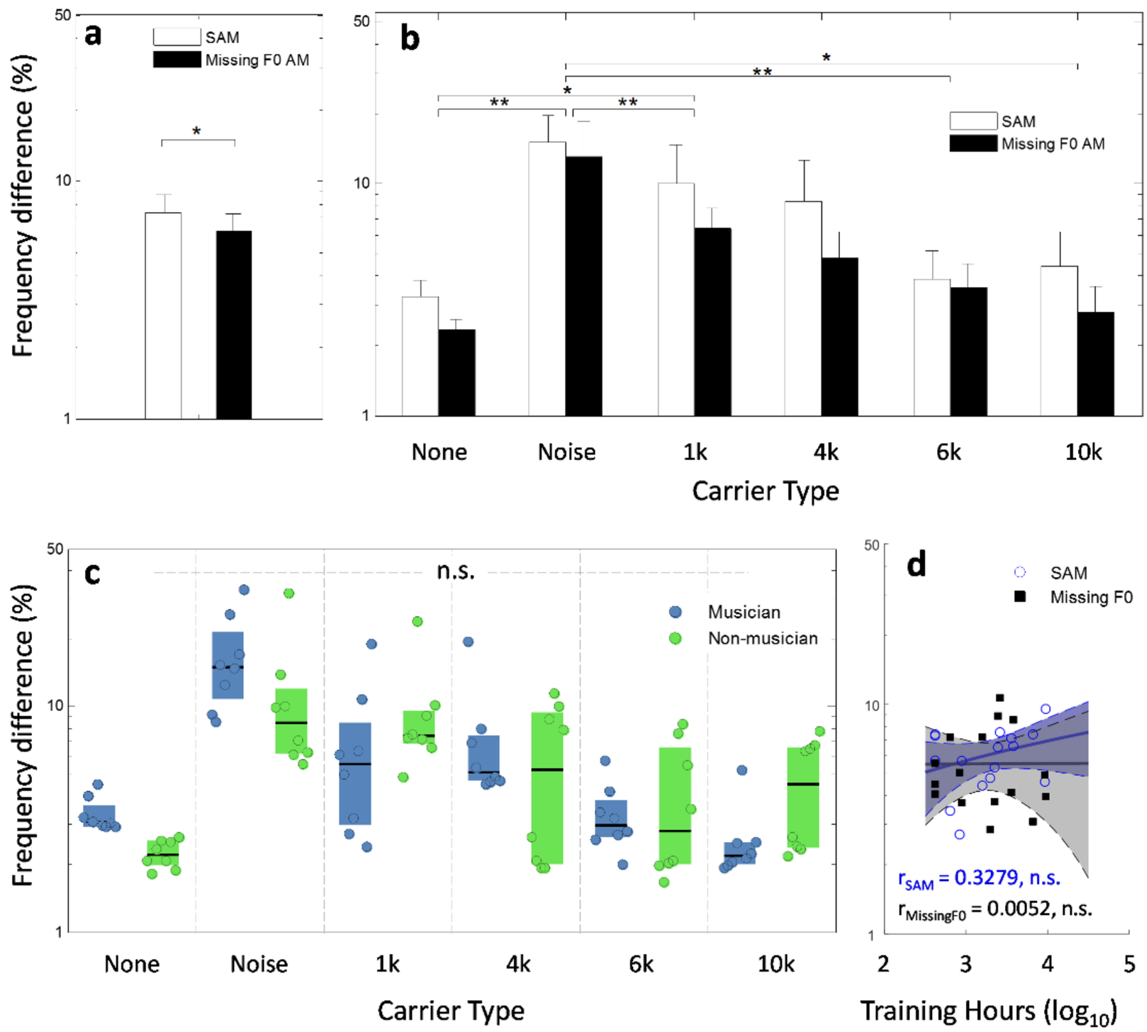

Fig. 2 Frequency difference limen results of F0-pitch tracking of Experiment 1. **a** Mean frequency difference limen in percentage (log scale) as a function of envelope type. **b** Frequency difference limen as a function of envelope and carrier type. **c** Frequency difference limen as a function of musicianship and carrier type. **d** Correlations between participant music training experience and frequency difference limen as a function of carrier type. Lines represent simple linear fit to the data. Shaded area represents 95% CI. Error bars represents ±1 standard deviation. n.s. = not significant, $*p < .05$, $**p < .01$. (Color figure online)

and continued to take formal music lessons for more than 7 years (>7 years of musical activity and >4,000 hrs of cumulative lifetime practice). The nonmusician group was defined as individuals with less than 1 year of experience with formal music lessons (except for compulsory music courses in school). This division resulted in eight participants in the musician group and eight participants in the nonmusician group. No musician advantage was observed in F0 pitch discrimination for any carrier type, $F(1, 14) = 0.201$, $p = .660$ (Fig. 2c). Pearson correlation tests revealed no significant correlation between hours of musical training received and F0 pitch thresholds for transposed tones modulated with SAM and missing-F0 envelopes, respectively ($r = .328$, $p = .107$ and $r = .005$, $p = .493$; Fig. 2d).

Experiment 2: Musical-pitch interval discrimination with transposed tones

In Experiment 1, we demonstrated that listeners could discriminate fine pitch differences based on the modulation rates of the SAM or missing-F0 envelopes of transposed tone complexes. In Experiment 2, we assessed the effects of carrier spectral region to musical-pitch intervals discrimination whose interval size information is conveyed by the temporal envelopes or SAM or the upper three-harmonics of identical F0s. A musical-pitch interval discrimination paradigm was used here to avoid the musical nature of directly naming the musical interval (which requires musical expertise). Pitch interval discrimination accuracy with transposed tones was expected to differ for low vs. high (> 4k Hz) carrier frequencies.

Method

Participants

A total of 16 participants (nine females) with a mean age of 21.7 years ($SD = 2.35$) years who had not taken part in Experiment 1 participated. All participants were normal-hearing, native speakers of Mandarin Chinese. All other requirements were identical as in Experiment 1.

Stimuli

The stimuli in each trial consisted of four sequential transposed tones grouped into two melodic pitch intervals. Interval 1 consists of tones A and B and Interval 2 consists of tones C and D (Fig. 3a). All tones were 500 ms in duration, including 20-ms linear rise and decay ramps. The two tones within each interval was separated by a 60-ms silence gap. The silence between two melodic-pitch intervals was 600 ms. Each trial consisted of ascending intervals (i.e., the second tone in each interval was always higher in frequency than the first) with either the same starting or ending note. The two intervals were designed to be unequal in magnitude with the following constraints. The sizes of the two intervals were designated i and i+Δi, where i is the magnitude of the smaller standard interval and i+Δi is the magnitude of the comparison interval. The size of i was randomly selected from 1, 3, or 6 semitones and the size of Δi fixed at 4 semitones (Luo et al., 2014; McClaskey, 2017). The starting (or ending) note was randomly roved over a continuous range of ±0.5 octave from its center frequency that was randomly

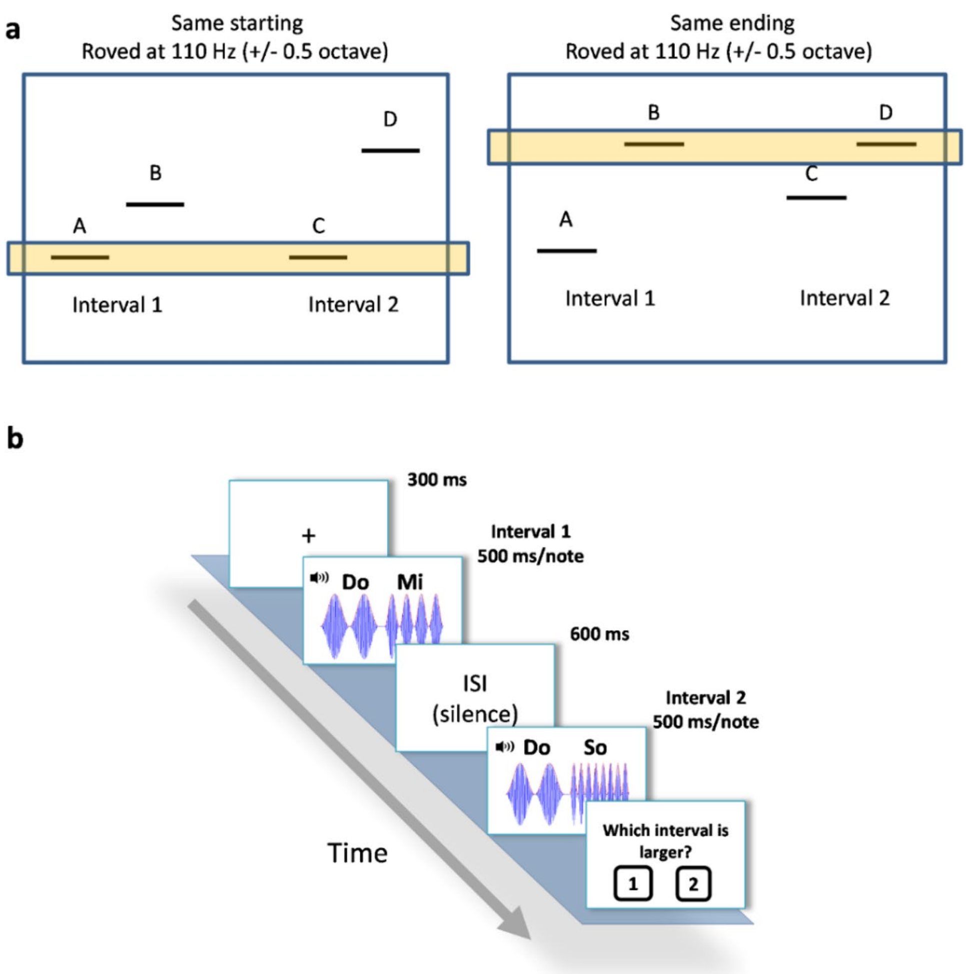

Fig. 3 Musical-interval discrimination paradigm of Experiment 2. **a** Each interval comprised two transposed tones with SAM or upper-harmonic envelopes with F0 corresponding to musical-note frequency modulating on high-frequency carriers. The starting or ending frequency within the two intervals were identical and roved with respect to a center frequency of 110 Hz ± 0.5 octave. **b** Diagram of the musical-interval discrimination task procedure. (Color figure online)

chosen in the 3rd octave frequency range (i.e., C3–B3; 130.8 Hz–246.9 Hz). The variation in starting (or ending) note prevented listeners from performing the task by simply choosing the interval with the highest or lowest frequency note. The order of presentation of the two intervals was counterbalanced across trials.

Procedure

The experiment was conducted using a block design. In a given block, one combination of envelope condition (SAM or missing-F0) and carrier type (none, 1k, 4k, 6k, 10k Hz and low-pass noise at 12 k Hz) was randomly selected given a total of 12 blocks. Within each block, the three different interval sizes (Δi = 1, 3, or 6 semitones) were intermixed within trials yielding a total of 300 trials (given approximately 100 trials per interval-size condition). The 12 blocks of envelope and carrier conditions were randomly selected, without replacement, until a full set of 12 blocks was completed. A two-interval forced-choice (2IFC) paradigm was used to assess pitch-interval judgment. On each trial, participants were presented binaurally through headphones with the two melodic-pitch intervals separated by a 1,000-ms ISI. Participants were instructed to choose whether the first or second interval contained the "larger" interval of the two with a key-press response. No feedback was given. Participants were permitted to take a short 5-min break between blocks as needed. Each participant completed a total of 3,600 trials (12 blocks × 300 trials/block). Figure 3b shows a schematic procedure of the pitch interval discrimination task.

Results

Carrier frequency affects musical-pitch interval discrimination with transposed tones

Figure 4 summarizes the results for pitch-interval discrimination accuracy across carrier frequencies. A two-way repeated-measures ANOVA was performed to evaluate the effects of envelope (SAM vs. missing-F0) and carrier type (none, noise, 1k, 4k, 6k, and 10k Hz) combined across interval sizes on the accuracy of pitch-interval discrimination. Interval-discrimination accuracy was significantly higher when the interval tones were conveyed via envelope modulations of three-harmonics compared with SAM with identical F0, $F(1, 15) = 12.00$, $p = .003$ (Fig. 4a). There was a significant main effect of carrier type on accuracy of pitch-interval discrimination, $F(5, 75) = 42.485$, $p < .001$ (Fig. 4b). The analysis also revealed a significant interaction between envelope and carrier type on discrimination accuracy of relative-pitch intervals, $F(5, 75) = 21.2789$, $p < .001$ (Fig. 4b). Post hoc analysis using Bonferroni corrections on carrier type showed that the no carrier condition (i.e., envelope only) resulted in higher pitch-interval discrimination accuracy than the noise and all levels of sinusoidal carriers (all ts > 3.25, ps < .01; details in Supplementary Table 1). When the envelope conveying pitch interval information was on the noise carrier, discrimination accuracy was significantly lower compared with all levels of sinusoidal carriers except for the 1k Hz carrier (all ts > −5.719, ps < .001; details in Supplementary Table 1). A linear trend contrast analysis on sinusoidal carriers showed no downward trend on pitch-interval performance with increasing carrier frequency, $F(1, 15) = 3.970$, $p = .056$.

Standard-interval size affects musical-interval discrimination

Next, the effect of standard-interval size on pitch-interval discrimination accuracy was examined. Figure 4c shows performance accuracy in pitch interval discrimination as a function of the size of standard interval combining across all carrier conditions. A two-way repeated-measures ANOVA was performed to evaluate the effects of envelope (SAM vs. missing-F0) and standard-interval size (1, 3, and 6 semitones) on the accuracy of pitch-interval discrimination. Accuracy of pitch-interval discrimination was significantly higher for tones conveyed via the envelope modulations of three-harmonics envelopes compared with SAM envelopes, $F(1, 15) = 29.135$, $p < .001$ (Fig. 4c). Performance accuracy of pitch-interval discrimination differed significantly as a function of pitch interval size, $F(2, 30) = 7.67$, $p = .002$ (Fig. 4c). No significant interaction between envelope type and interval size on discrimination accuracy of pitch intervals was obtained, $F(2, 30) = 2.749$, $p = .080$. Post hoc analysis on the size of standard-interval revealed a significant difference in discrimination accuracy between the 1- and 6-semitone interval conditions, $t(31) = 4.063$, $p < .001$, and the 3- and 6-semitone interval conditions, $t(31) = 3.118$, $p = .004$ (Fig. 4c). Specifically, larger standard interval size resulted in markedly poorer performance compared with smaller standard-interval size.

Musical-interval discrimination with SAM-envelope modulated transposed tones

For tones modulated with SAM envelopes, the effects of carrier type and interval size on relative-pitch discrimination accuracy were assessed with a two-way repeated-measures ANOVA. The analysis revealed a significant main effect of carrier type on pitch-interval discrimination accuracy, $F(5, 75) = 45.283$, $p < .001$ (Fig. 4d, left panel). There was a significant main effect of standard-interval size on discrimination accuracy, $F(2, 30) = 7.467$, $p = .002$. A significant interaction between carrier type and interval size

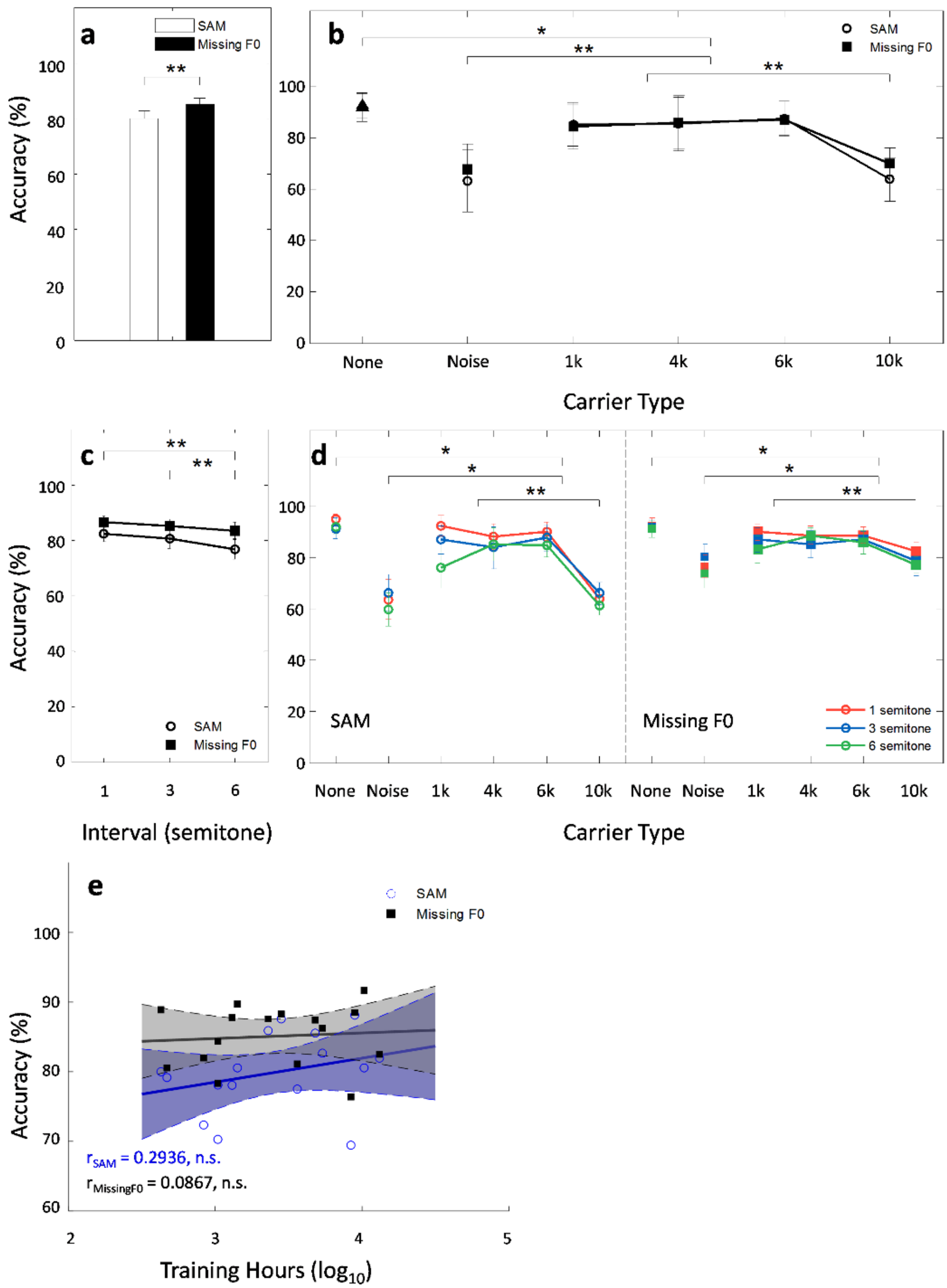

Fig. 4 Musical interval discrimination accuracy results of Experiment 2. **a** Mean accuracy in musical-interval discrimination as a function of envelope type. **b** Mean musical-interval discrimination accuracy as a function of carrier and envelope type. **c** Mean accuracy as a function of interval size in semitone. **d** plotted as a function of different type of carriers for SAM envelope (left panel) and missing-F0 envelope (right panel). **e** Correlation between musical training hours and pitch-interval discrimination accuracy for transposed tones modulated with SAM (blue circles) and three-harmonic (black squares) envelopes. Lines represent simple linear fit to the data. Shaded area represents 95% CI. Error bars represents ±1 standard deviation. n.s. = not significant, *$p < .05$, **$p < .01$. (Color figure online)

on discrimination accuracy of pitch intervals was obtained, $F(10, 150) = 2.742$, $p = .004$. Post hoc analyses revealed significantly higher pitch-interval discrimination accuracy when the pitch was conveyed solely by the envelope (i.e., no carrier) compared with when it was carried on either noise or any of the sinusoidal carriers (all *t*s > 3.265, *p*s < .005; details in Supplementary Table 2). Compared with noise carrier, envelopes imposed onto sinusoidal carriers produced better performance on pitch-interval discrimination, $t(15) = -8.616$, $p < .001$, $t(15) = -9.361$, $p < .001$, and $t(15) = -10.787$, $p < .001$ for 1k, 4k, and 6k Hz carriers, respectively. All pair-wise comparisons between sinusoidal carriers were not significant except for the comparison between the 1k Hz carrier and all other higher-frequency carriers (all *t*s > 6.676, *p*s < .001; details in Supplementary Table 2).

Musical-interval discrimination with missing-F0 envelope modulated transposed tones

For tones modulated with envelope of the three-harmonics, a two-way repeated-measures ANOVA, with the factors carrier type and interval size, revealed a significant main effect of carrier type on the accuracy of pitch-interval discrimination, $F(5, 75) = 19.428$, $p < .001$ (Fig. 4d, right panel). There was a significant main effect of interval size on pitch-interval discrimination accuracy, $F(2, 30) = 4.443$, $p = .020$. No significant interaction between carrier type and interval size on interval discrimination accuracy was observed, $F(10, 150) = 1.868$, $p = .054$. Post hoc comparisons on carriers showed that the envelope only condition (i.e., no carrier) resulted in a significantly higher pitch-interval discrimination accuracy compared with all other carriers (all *t*s > 3.179, *p*s < .006; details in Supplementary Table 3). Except for the 10k carriers, all other sinusoidal carriers resulted in better pitch-interval discrimination accuracy compared with the noise carrier, $t(15) = -5.745$, $p < .001$, $t(15) = -5.484$, $p < .001$, and $t(15) = -6.764$, $p < .001$ for the 1k, 4k and 6k Hz carriers, respectively. Pair-wise comparisons between sinusoidal carriers showed a significant difference between the 10k Hz carriers and the lower-frequency carriers, $t(15) = 4.385$, $p < .001$, $t(15) = 5.317$, $p < .001$, and $t(15) = 7.485$, $p < .001$ for the 1k, 4k and 6k Hz carriers, respectively.

Musical training effects on pitch-interval discrimination

Lastly, a Pearson correlation analysis was performed between music training and pitch-interval discrimination accuracy. We obtained the total lifetime practice hours for each participant from the MMHQ as an index for musical training experience. No significant correlation between musical training hours received and pitch-interval discrimination accuracy was obtained for either type of envelope modulations, $r = .294$, $p = .135$ and $r = .087$, $p = .375$ (Fig. 4e) for SAM and missing-F0 envelope conditions, respectively.

Experiment 3: Melody identification with transposed tones

Experiment 3 examined the ability to use temporal information conveyed by envelope modulation of transposed tones to derive the pitch contour for melodic processing under different carrier spectral regions. Melodic sequences were conveyed in the rates of temporal envelope modulations of SAM or upper-harmonic complex envelopes transposed to different carrier spectral regions. Based on previous results, we expected melodic identification based on the envelope periodicity of transposed tones to decline with increasing carrier frequency.

Method

Participants

A total of sixteen participants (10 females) with a mean age of 22.06 years ($SD = 1.85$) years who had not taken parts in Experiments 1 and 2 participated. Participants fulfilled the same requirements as in Experiment 1.

Stimuli

The stimuli were a set of 15 (±2)-note melodic sequences where each note of the melodic sequence was conveyed by the envelope modulation rate corresponding to its musical note frequency and transposed onto different carrier types following the same stimulus procedure in Experiment 1. The original unaltered melody excerpts were selected from familiar children's songs or nursery rhymes in Mandarin Chinese familiar to most people growing up in Taiwan. To minimize rhythmic cues, the original version of the 13 selected melodies were modified in MATLAB to consist of equal-duration pure tones at 500 ms with an intertone interval of 250-ms silence. A linear 10 ms rise-decay ramp was imposed on each note to reduce spectral splatter. The frequency range of the original musical excerpts was limited to within one octave range on Western music convention (i.e., C3–B3; 130.8 Hz–246.9 Hz) and contained only white-key notes. Example of the song selection include "Twinkle, Twinkle, Little Star" and "Home, Sweet Home" (details in Supplementary Table 4). Following the same procedure, the transposed tone version of the melodies contained either SAM or missing-F0 envelope modulation multiplied onto carriers at 1k, 4k, 6k, 10k Hz and a low-pass noise at 12k Hz. For comparison, a no carrier condition (i.e., envelope only) where the melody was played by either varying the modulation rate of the SAM or missing-F0 harmonic envelopes was also included.

Procedure

The experiment was run in a randomized block design. In a given block, one of the 10 stimulus conditions (2 envelopes × 6 carrier types) was randomly selected. Each block consisted of 15 trials. For each session, the 12 blocks of envelope-type and carrier combination conditions were randomly selected, without replacement, until a full set of 12 blocks was completed. Each participant completed two sessions of 12 blocks each. On each trial, a randomly selected melody of a given condition was presented twice in succession with an intertrial interval of 1,000 ms. A single-interval, four alternative forced choice paradigm was used to test melody identification performance. Subjects were asked to indicate which one of the melodies was heard by choosing from one of the four options of melody titles presented as GUI on the screen. No feedback was given. Participants were allowed to take short 5-min breaks between blocks at any time. Each participant completed a total of 360 trials (2 sessions × 12 blocks/session × 15 trials/block). Prior to the experiment, participants were required to pass a pretest consisting of 10 melody-identification questions with an accuracy of 90% or above to make sure that they were familiar with the original (without carriers) melodies. The pretest melodies were excluded from the actual experiment task. The experimental task took approximately 50 min to complete.

Results

Melody identification with transposed tones

Figure 5 summarizes the performance in melody identification with transposed tones. Figure 5a shows melody identification accuracy as a function of envelope modulation combined across all carrier types. Melody identification accuracy was significantly higher when the tones were modulated with SAM envelope compared with envelope of the upper-harmonics, $t(15) = 8.556$, $p < .001$. Figure 5b shows melody identification accuracy with transposed tones as a function of envelope and carrier type. A two-way repeated-measures ANOVA was performed on melody- identification accuracy with the factors envelope type (SAM vs. missing-F0) and carrier (none, noise, 1k, 4k, 6k, and 10k Hz). The analysis resulted in a significant main effect of envelope type on accuracy of melody identification, $F(1, 15) = 73.204$, $p < .001$. Melody identification accuracy was greater when the transposed tones were modulated with SAM envelope compared with those of missing-F0 envelope conditions. There was a significant main effect of carrier type on accuracy of melody identification, $F(5, 75) = 21.665$, $p < .001$. The analysis resulted in a significant interaction between envelope type and carrier on melody identification accuracy, $F(5, 75) = 57.913$, $p < .001$. Post hoc comparisons of carrier type showed that melody identification accuracy was lower when the envelope conveying melody was carried by noise compared with when it was conveyed by sinusoidal carriers at frequencies below 6 kHz, $t(15) = -10.489$, $p < .001$, $t(15) = -16.441$, $p < .001$, and $t(15) = -10.470$, $p < .001$ for 1k, 4k, and 6k Hz carriers, respectively. Compared with the envelope only condition (i.e., no carrier), identification performance was only markedly poorer when the melody was carried by the noise carrier, $t(15) = 15.605$, $p < .001$ and the 10k Hz carrier, $t(15) = 3.605$, $p = .003$. Melody identification accuracy was significantly lower when the melody was carried by the highest-frequency carrier (i.e., 10k Hz) compared with all sinusoidal carriers below 6k Hz—1k: $t(15) = 3.931$, $p < .001$; 4k: $t(15) = 4.759$, $p < .001$; 6k: $t(15) = 3.808$, $p = .003$. All other post hoc comparisons were not significant. A linear trend contrast on sinusoidal carriers revealed no downward trend on melody identification performance with increasing carrier frequency, $F(1, 15) = 1.126$, $p = .305$.

Musical training effects on melody identification with transposed tones

A Pearson correlation analysis between music training hours and melody identification accuracy was performed to assess whether music training experience benefits identification of melodies conveyed via transposed tones. No significant correlation between hours of musical training received and melody identification accuracy was obtained for transposed tones modulated with either SAM ($r = -.259$, $p = .332$; Fig. 5c) or missing-F0 envelopes ($r = -.047$, $p = .862$; Fig. 5c).

Simulation with a model of the auditory periphery

The peripheral auditory response to transposed tones with a 110-Hz modulation imposed onto 1k Hz and 6k Hz carriers were simulated in MATLAB using a bank of fourth-order GammaTone bandpass filters logarithmically spaced between 50 and 8000 Hz (Holdsworth et al., 1988), with filter bandwidths based on human auditory filter estimates measured in notched-noise (Glasberg & Moore, 1990). Signal output within each channel of filter bank was weighted by a frequency-dependent function representing outer- and middle-ear attenuation derived from Meddis and Hewitt 1991a, b), followed by autocorrelation within frequency channels and frequency integration across channels (Hsieh & Saberi, 2016). The peak of the combined correlation

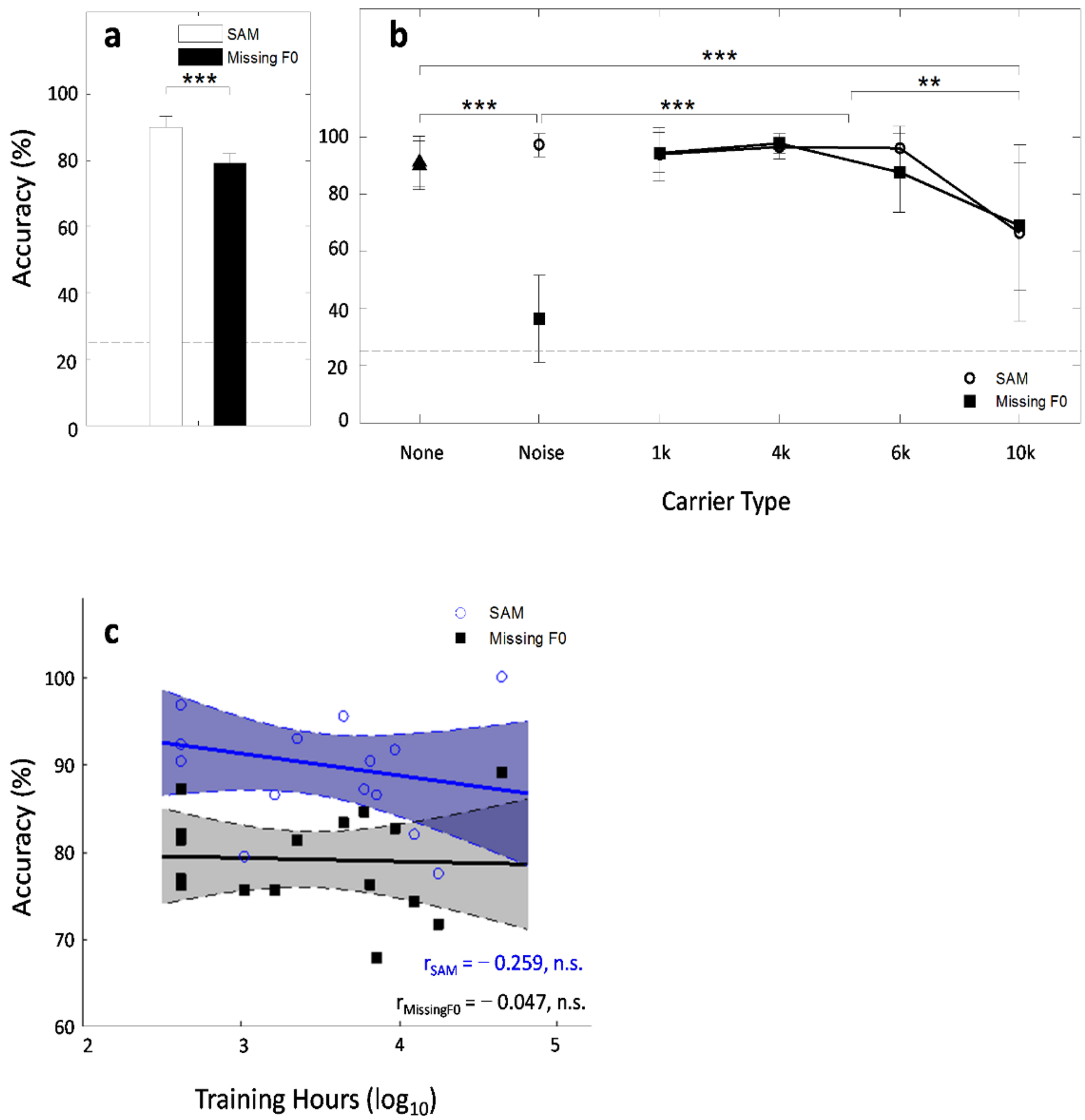

Fig. 5 Melody identification accuracy results of Experiment 3. **a** Melody identification accuracy as a function of envelope type combined across all carriers. **b** Mean accuracy as a function of carrier and envelope type. Dashed line indicates change performance. **c** Correlation between music experience (training hours) and melody-identification accuracy for tones modulated with SAM (blue circles) and missing-F0 envelopes (black squares). Shaded area represents 95% CI. Error bars represents ±1 standard deviation. n.s. = not significant, *$p < .05$, **$p < .01$, ***$p < .001$. (Color figure online)

function reflects the common pitch. Figure 6a–d shows the time waveform representations for SAM and missing-F0 envelope modulated transposed stimuli on 1k and 6k Hz sinusoidal carriers compared with a noise carrier and envelope only conditions. Figure 6e–h shows the corresponding Fourier frequency spectrum. Figure 6i–l shows the output of this model prior to frequency integration for transposed tones with sinusoidal carriers of 1k, 6k Hz, a noise carrier and pure envelope modulation without a carrier. Figure 6m–p shows the outputs after integration across frequency channels. Transposed tones with either SAM or missing-F0 modulated envelopes on sinusoidal carriers show a clear peak at 9-ms lag corresponding to the reciprocal of its envelope modulation rate of 110 Hz (Fig. 6i–j). In comparison, the autocorrelation output of the SAM noise exhibits no clear peaks at the 9-ms lag (Fig. 6k). Further, comparing simulated filter response indicates a narrower peak associated with tones modulated with a missing-F0 envelope compared with those modulated with the SAM envelope. This suggests that sufficient pitch information, comparable to that of the envelope-only condition, is available in the autocorrelation functions of transposed tones for pitch discrimination.

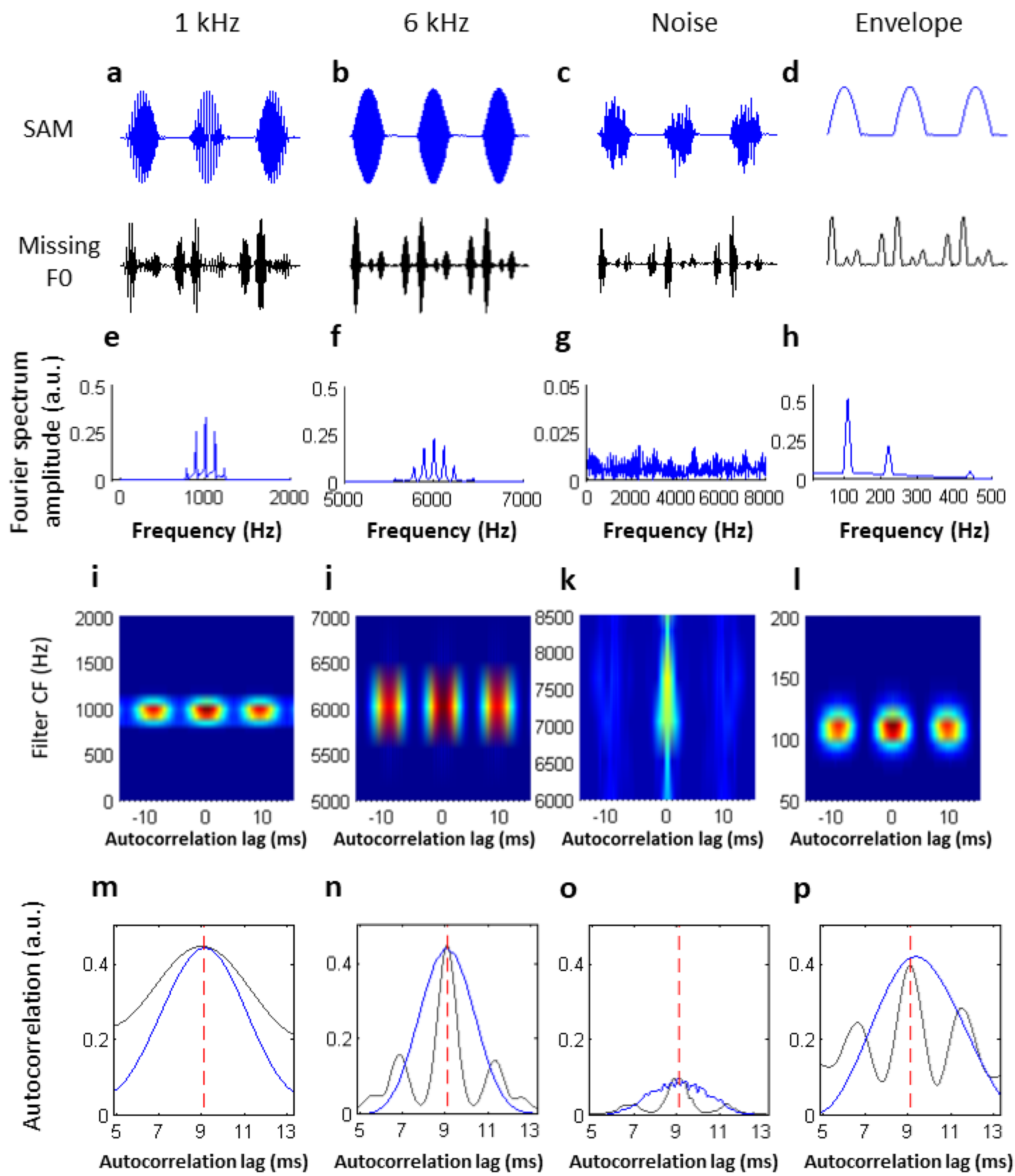

Fig. 6 Output of an autocorrelation model of the auditory periphery. Top panels show time waveforms and bottom two panels show their autocorrelation functions respectively. **a–d** Time waveforms for a 110 Hz SAM (blue) or missing-F0 (black) envelope modulating 1k Hz, 6k Hz, noise carriers, and envelope only (no carrier) conditions, respectively. **e–h** Fourier frequency spectrum corresponding to the 110 Hz SAM envelope modulating 1k Hz, 6k Hz, noise carriers, and envelope only conditions. **i–l** Model output prior to integration across frequency channels. **m–p** Model output after integrating across frequency channels. Red vertical dashed line indicates the expected time lag corresponding to an envelope modulation at 110 Hz. Both the SAM and missing-F0 envelope transposed tones produced a peak at a time lag of 9 ms equivalent to those produced by the pure sinusoidal envelope condition. (Color figure online)

Discussion

Using transposed tones to convey identical temporal envelope modulation patterns across carrier spectral regions, we showed that the extraction of musical pitch from temporal envelope information can be reliably achieved across different spectral regions. No musician advantage was found in musical pitch extraction from the envelope (SAM or missing-F0 harmonic complex) of modulated sounds across different spectral regions. Experiment 1 showed that overall pitch discrimination thresholds on transposed tones were below 5% with lower thresholds associated with increasing carrier frequency. Importantly, comparable thresholds of pure sinusoids and transposed tones on high-frequency carriers were observed. Experiment 2 showed that listeners can discriminate musical pitch intervals defined by the modulation rate of the SAM or missing-F0 harmonic complex envelopes, even when the carrier frequency exceeds 6k Hz. Additionally, the magnitude effect of standard-interval size in pitch-interval discrimination associated with pure sinusoids was observed with transposed tones. Experiment 3 showed that melodic contours can be accurately conveyed by the periodicity of the SAM or harmonic-complex envelopes across carrier frequencies up to 6k Hz, with a slight drop but above chance performance at 10k Hz. Overall, these findings suggest that the information extracted from temporal envelopes across carrier tonotopy can provide sufficient frequency information to allow highly accurate F0 pitch discrimination and melody/interval identification.

Importantly, our results suggest a perceptual enhancement for temporal envelope information based on different carriers, with better musical pitch perception observed at higher-frequency carriers up to 6k Hz. For F0 pitch discrimination, an enhanced pitch discrimination sensitivity (i.e., lower-frequency difference thresholds) was observed with increasing carrier frequency. This suggests that higher-frequency carriers may actually enhance the temporal envelope information for pitch extraction, and highlights the importance of cochlear place coding (Whiteford et al., 2020). We discuss here three potential reasons why high-frequency carriers improved pitch perception of AM envelopes compared with low-frequency ones. One possibility is that the spectral separation between carriers and envelope is greater for higher-frequency carriers (6k, 10k) compared with carriers in lower-frequency regions of the cochlea (i.e., 1k, 4k), hence producing less interference to the extraction of envelope periodicity. Consistent with this, previous studies have reported a greater interference effect to modulation frequency discrimination when the sinusoidal target and interferer were presented at the same spectral location, and when the interferer modulation was lower in frequency than the target modulation (Carlyon, 1996; Gockel et al., 2005; Kreft et al., 2013; Mehta & Oxenham, 2018). A second possibility might be due to the lack of competing temporal cues for transposed tones with carriers above 4k Hz (i.e., both temporal and spectral information are available below 4k Hz). Specifically, low-frequency sounds (below 4k Hz) convey pitch percepts via both spectral and temporal cues, and as such, temporal cues related to carriers (per se) might interfere with envelope temporal cues for deriving pitch percepts. This is different from high-frequency carriers where only spectral cues are available, thus no conflicting temporal cues exist between carriers and its AM envelope for pitch perception. Lastly, the output of hair-cell rectification and modulation extraction process differs for high versus low-frequency AN fibers. Given that the tuning filter and low-pass filter only overlap in low-frequency carriers, the hair cell output contains both the modulation and carrier information in low-frequency AN fibers, whereas the output of high-frequency AN fibers strictly contains the modulated envelope. Consequently, the same amplitude envelope is better represented in high-frequency relative to low-frequency AN fibers, which may be why high-frequency carriers facilitated pitch perception of AM envelopes compared with low-frequency ones.

Our findings implicate the efficacy of temporal envelope cues to pitch perception on high-frequency sounds may be more pronounced than previously considered. This is consistent with previous reports that temporal envelope modulations, represented as amplitude modulations on high-frequency AN fibers, provide sufficient information to facilitate accurate pitch discrimination for transposed single tones (Oxenham et al., 2004) and precise interaural time difference cues for binaural lateralization (Bernstein & Trahiotis, 2002, 2011; Monaghan et al., 2015). In addition, the use of temporal envelope cues in high-frequency regions corroborates previous reports that pitch percepts evoked by complex tones with higher unresolved harmonics are sufficient for melody recognition, consonant/dissonant interval discrimination, and detection of mistuned harmonic components (Carcagno et al., 2019; Gockel & Carlyon, 2018; Moore & Sęk, 2009; Oxenham et al., 2011). It has been suggested that the pitch induced by harmonic complexes containing only unresolved harmonics in high-frequency regions (i.e., >4k Hz) is primarily based on envelope periodicity/phase locking, and is sensitive to the salience of temporal envelope cues (Plack & Oxenham, 2005a, b; Song et al., 2016). In addition, predictions simulated using an autocorrelation model of pitch extraction (Glasberg & Moore, 1990; Holdsworth et al., 1988; Meddis & Hewitt, 1991a, b) showed a clear peak at the F0s of either pure sinusoids or the three-harmonic envelopes (Fig. 6) that corresponds to perceived pitches, consistent with our current observations. Such temporal models would tend to disregard the tonotopic organization of the cochlea in pitch computation. Note that although the present results, in general, agree with the temporal based accounts of pitch perception, the contribution from carrier tonotopic information should not be disregarded.

Compared across carrier types, the best musical pitch performance was associated with the envelope only condition (i.e., no carrier), followed by sinusoidal carriers, with the worst performance for noise carriers. This could be attributed to the fact that spectral side bands effects on AM detection thresholds are more pronounced on sinusoidal carriers relative to noise carriers (Kohlrausch et al., 2000; Moore & Glasberg, 2001). In fact, AM detection on modulated noise has been thought to rely only on temporal resolution per se since modulation of white noise does not introduce any change to its magnitude spectrum (Moore & Glasberg, 2001). Compared with sinusoidal carriers, the lower autocorrelation outputs at the periodicity corresponding to the F0 of perceived pitches on noise carriers (Fig. 6k) corroborates the decline in performance observed on noise carriers. In addition, it has been suggested that the inherent amplitude fluctuations in a noise carrier can limit the detectability of its imposed modulation (Dau et al., 1999). Apart from noise carriers, the perceptual enhancement of temporal envelope information also seems to differ for carriers belonging to the EHF region (i.e., 8–20k Hz). Specifically, we note a slight drop in performance associated with the 10k Hz carrier for musical interval and melody identification tasks but not for F0 discrimination. In the musical-interval comparison and melody-identification task, it requires not only the ability to differentiate high or low tone, but also the ability to judge the distance between two tones which may also involve working memory. Additionally, the performance difference due to task requirement may be more pronounced for less familiar sounds (i.e., 10k Hz) for which a change in timbre could be detected. In fact, timbre is one factor that has been shown to affect the perceived size of melodic intervals (Russo & Thompson, 2005; Zarate et al., 2013), but not pitch threshold detection (Holmes et al., 2022). This may partially explain why the perceptual enhancement of temporal envelope information was more pronounced for F0 pitch discrimination compared with musical-interval or melody identification for carriers in the EHF region (i.e., 10k Hz).

Some studies have demonstrated a potential musician benefit in F0-pitch or pitch-interval discrimination tasks, suggesting that musicians may use a different strategy (such as different use of phase-locking cues or more reliance on phase-locking cues) in extracting or processing pitch-related information (Madsen et al., 2017; McClaskey, 2017; Micheyl et al., 2006). To examine the potential factor of pitch-processing advantage due to musical expertise, correlation analysis between musical training and performance was performed for all three pitch tasks. No correlation was observed between music training hours and pitch thresholds, musical-interval or melody identification performance, suggesting that temporal-envelope based pitch encoding across different spectral regions may not be affected by musical experience. This is consistent with McClaskey's (2017) findings as they did not observe a pitch-interval perception advantage associated with amateur musicians. However, it is contrary to studies that have reported musicians to exhibit more precise phase-locked brainstem frequency following response (FFR) to temporal envelope periodicities when discriminating musical pitch and musical intervals compared with those of nonmusicians (Bidelman et al., 2011; Lee et al., 2009; Musacchia et al., 2008). One speculation for the lack of musician benefits in pitch perception may be due to the use of different stimuli. All prior studies that observed musician effects on pitch perception have typically used pure tones (Kishon-Rabin et al., 2001) or harmonic complex tones containing resolved harmonics (Allen & Oxenham, 2014; Micheyl et al., 2006). Here, we used musical-note frequency (as envelopes) to modulate low and high-frequency carriers as stimuli (i.e., transposed tones), which may contribute to the difference in performance. The unfamiliarity of transposed tones in music training may further contribute to the lack of musician effects. Second, not all pitch-related ability is trainable or detectable. For instance, Bianchi et al. (2016) has shown that musicians and nonmusicians share similar peripheral frequency selectivity in detecting unresolved harmonic complexes. Besides, the "musician pitch enhancement" has been hypothesized to occur at the cortical level of the auditory system, and is detected better with neuroimaging tasks or tasks related to processing efforts (Bianchi et al., 2016). Lastly, the sample size and selection criteria for musicians may have limited the interpretation of the lack of musician effects on pitch perception across spectral regions. Indeed, the amount of training and selection criteria of musicians has been shown to affect the extent of pitch enhancement (Micheyl et al., 2006). Future studies can incorporate more stringent criteria in selecting musicians with professional training to see whether there might be a musician advantage in processing temporal envelope and/or fine structure information in musical pitch perception.

From a biological standpoint, the capabilities to encode the envelope information in high-frequency regions is essential to auditory perception and has ecological relevance, since a multitude of important information is conveyed in temporal envelope modulations of naturally occurring sounds. That the temporal envelope remains a robust cue when the conveying carrier information is degraded or presented in different spectral regions may be perceptually relevant for the survival of species under complex listening conditions. In agreement with this viewpoint, studies on nonhuman primates and other mammals have also demonstrated sound periodicity detection through temporal, but not spectral processing (Shofner & Chaney, 2013; Song et al., 2016). On a practical level, the fact that pitch information conveyed via identical temporal envelopes is perceived differently at different cochlear locations might provide potential implications for cochlear implant (CI) processing. This finding corroborates previous studies that highlight the importance of spectral information for music perception in CIs (Kong et al., 2004). Moreover, we used the missing-F0 envelope condition here to simulate quasi-stochastic temporal firing activities in the corresponding auditory nerve fibers. The finding that complex-harmonic envelopes (i.e., more stochasticity) outperformed SAM envelopes across all carrier types in both pitch discrimination and pitch-interval identification suggests a potential stochasticity-induced advantage to pitch perception. This is consistent with Rubinstein et al. (1999) who showed that applying high-rate background pulses to introduce more stochastic activity in the corresponding auditory nerve fibers can enhance the temporal representation of modulated signals. Such advantage was not observed in the melody-identification task. Rather, we observed that extracting the temporal periodicity corresponding to the F0s (of three-harmonic envelopes) on noise carriers was more difficult in melody identification, which seems to echo the poor melody perception reported for CI users (McDermott, 2004). Even so, our study demonstrates that pitch perception could be strengthened by manipulating the envelope information conveyed on a broadband noise or via an electrode of CI. Future studies can examine whether single electrode firing can convey more pitch information to CI users. In addition to those with severe hearing loss, manipulating the temporal envelope cues naturally or by artificial means can benefit those with poor cognition and auditory neuropathy spectrum disorder. As such, these results suggest that enhancing temporal envelope cues at certain carrier frequency ranges (i.e., electrodes) may be beneficial for musical pitch perception and appreciation in CI and other related areas (Vandali et al., 2005).

Conclusion

In summary, our results showed that the temporal envelope modulation characteristic of natural sounds across tonotopic regions of the cochlea provides sufficient information for reliable pitch discrimination and melodic-pitch processing. For certain modulation frequencies, enhanced pitch discrimination associated with the sine or complex-harmonic envelopes was observed at high-frequency carriers, with performance comparable to those of pure sinusoids. Pitch interval and melody identification did not decline with increasing carrier frequency, suggesting that pitch perception based on temporal envelope cues is not constrained by the spectral region of carriers. Findings suggest that the upper limit of temporal envelope cues in encoding the (musical) pitch of a high-frequency sound may be much higher than previously considered, with implications for supporting ecologically relevant auditory functions.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.3758/s13414-023-02761-x.

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Open practice statements The data and materials for all experiments are available upon request to the corresponding author. None of the experiments was preregistered.

Authors' contributions C.K., C.J., and I.H. designed the experiments. C.K., J.L., and I.H. programmed the auditory stimuli and experimental tasks. C.K. and J.L. conducted the experiments. I.H., C.J., and C.W. performed data analysis. C.K., J.L., and I.H. prepared the figures. All authors contributed to data interpretation and discussion. I.H. wrote the main manuscript text.

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Data Availability The data for this study is available upon request to the corresponding author.

Declarations

Ethics All experiments were approved by the Research Ethics Committee at National Taiwan University, Taiwan (NTU-REC No.: 202105EM029).

Competing interests The authors declare no competing interests.

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